

Solution Design Document

NutriTrack

Nutrition and Diabetes Awareness Analytics

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A self initiated personal project

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Version History

Version	Date	Author	Description
1.0	June 2026	Abhijith Nallana	Initial draft covering Phase 1: data pipeline, exploratory data analysis, and the Streamlit application.

Executive Summary

NutriTrack is a data application that brings together two trusted public sources, the USDA Foundation Foods nutrition database and the NHANES 2017 to 2018 national health survey, into one accessible tool. It allows a user to look up the nutrition of a food, log meals, track daily intake against personal goals, and explore how measurable factors such as glucose, body mass index, age, and physical activity relate to diabetes across a national sample.

This document describes the design of Phase 1, which delivers a complete and reproducible data pipeline, a full exploratory data analysis, and a working application. Phase 1 is descriptive and educational. It does not diagnose diabetes and does not apply a predictive model. The later phases extend the system with a machine learning classifier and a generative AI assistant, and they are summarized in the implementation timeline.

Purpose of Solution Design Document

The purpose of this document is to define the scope, design, and technical approach for the NutriTrack solution so that the work is clear, repeatable, and straightforward to evaluate. It records the business and functional requirements, the chosen technology stack, the flow of data through the system, the assumptions that shape the design, and the risks that need to be managed. It serves as the single reference that connects the goals of the project to the way the solution is actually built.

Project Background

Reliable nutrition and health data exists in the public domain, yet it is difficult for a non technical user to apply. The USDA Foundation Foods files and the NHANES survey are published as large raw files in different formats, one as comma separated values and the other as SAS transport files. Using them requires the skill to clean, join, and interpret thousands of records. As a result, the value held in these datasets rarely reaches the people who could benefit from it.

NutriTrack closes that gap. It transforms the raw public data into a clean and unified store, then presents it through a simple application. A user can search a food and read accurate nutrition for a chosen serving, record meals across a day, and view clear summaries of how common risk factors relate to diabetes in the survey population.

Project Team

NutriTrack is a self initiated personal project, designed and built from end to end by a single developer. There is no wider team. The project owner carries every role listed below.

- **Data engineering and analysis:** sourcing, cleaning, and preparing the USDA and NHANES data, and performing the exploratory analysis and diabetes insights.
- **Application development:** building the Streamlit application and the SQLite database.
- **Documentation:** preparing this design document and the project README.

Point of Contact

- **Project owner and developer:** Abhijith Nallana.

Business Requirements

The solution is expected to meet the following needs:

- Allow a user to search the food catalog and view accurate nutrition values scaled to a chosen serving size.
- Allow a user to log meals and track daily calorie and macronutrient totals against personal goals.
- Present clear and honest summaries of how factors such as glucose, body mass index, age, and physical activity relate to diabetes in the survey population.
- Keep the analysis transparent and reproducible, so that every figure can be traced back to the source data.

Scope

In scope for Phase 1 is the ingestion of the USDA Foundation Foods and NHANES 2017 to 2018 data, cleaning and feature preparation, a complete exploratory data analysis in a notebook, storage of the cleaned tables in a SQLite database, and a Streamlit application with six pages that reads from that database.

Out of scope for Phase 1 is any predictive or diagnostic model, real time data feeds, user accounts and authentication, and deployment to a public server. The predictive model is planned for Phase 2 and the generative AI assistant for Phase 3.

Success Criteria

Phase 1 is considered successful when the following are true:

- The pipeline runs from end to end and produces a clean SQLite database containing 328 USDA Foundation Foods and 9,254 NHANES respondents.
- The Streamlit application runs with a single command and serves all six pages using only the database.

- The analysis notebook demonstrates the required statistical concepts, and every output is accompanied by a written interpretation.
- The results are reproducible, so that re running the notebook regenerates the same database and the same findings.

Solution Overview

NutriTrack follows a clear separation between the data layer and the application layer. A Jupyter notebook performs the full extract, clean, and analyze workflow and writes two tidy tables to a SQLite database. The Streamlit application then reads that database in a read only manner and serves the user experience. Because the application never touches the raw files, the data work and the application work stay independent and easy to maintain.

The architecture is shown in Figure 1. Raw USDA and NHANES files enter the notebook, which cleans them and writes the foods and nhanes tables to SQLite. The application reads those tables and renders the six pages.

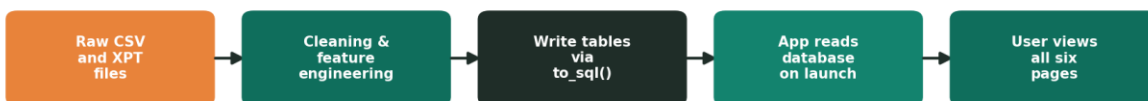
Figure 1. System Architecture



Automation Flow

The pipeline is automated from raw input to finished application. The notebook downloads the NHANES survey files automatically when they are absent, reads the USDA files that the user has placed in the data folder, applies all cleaning and feature steps, and writes the results to SQLite through a single call to the pandas to_sql function. The application reads the database when it launches and serves every page. Re running the notebook refreshes the entire database, so the system is simple to rebuild and the output stays consistent. Figure 2 shows this flow.

Figure 2. Automation Flow



Implementation Timeline

- **Phase 1, the data pipeline, exploratory analysis, and Streamlit application:** complete.
- **Phase 2, a diabetes prediction model on the cleaned survey table:** planned.

- **Phase 3, a generative AI nutrition assistant using retrieval augmented generation:** planned.

Technology Stack

- **Language and analysis:** Python, Pandas, and NumPy.
- **Visualization:** Plotly, Seaborn, and missingno.
- **Storage:** SQLite.
- **Application:** Streamlit.
- **Notebook environment:** Jupyter.
- **Planned additions:** scikit learn for the Phase 2 model and a retrieval augmented language model for the Phase 3 assistant.

Current State Process Flow

Today a user who wants to understand food nutrition and personal risk relies on scattered nutrition labels and general web searches. The authoritative public datasets exist, but they arrive as raw files in different formats and demand manual cleaning and joining before they can be read. The result is that useful information is hard to reach and easy to misread. Figure 3 illustrates this current state.

Figure 3. Current State Process Flow



Future State Process Flow

In the future state the user works inside one application. They search a food, log a meal, and immediately see their daily nutrition together with diabetes risk context, all served from a single cleaned database that was produced by an automated and reproducible pipeline. The friction of finding, cleaning, and interpreting raw data is removed. Figure 4 shows this target flow.

Figure 4. Future State Process Flow



Functional Requirements

- **Food search:** search the USDA catalog by name and display nutrition values scaled to a serving size that the user selects.
- **Meal logging:** record meals by date and meal type, and store the scaled nutrition for each entry.
- **Daily totals:** aggregate logged meals into daily calorie and macronutrient totals and compare them against the user goals.
- **Insight dashboards:** present diabetes prevalence by age group and by body mass index category, along with feature correlations.
- **Nutrition calculator:** estimate daily energy needs and present intake against goals.

Technical Requirements

- The notebook must run from start to finish without manual steps beyond placing the USDA files in the data folder.
- The cleaned data must persist in a SQLite database, with an index on food names for fast search.
- The application must read the database in a read only manner and launch with a single command.
- The data layer and the application layer must stay separate, so that a change in one does not break the other.
- The solution must run on a standard Windows environment inside a Python virtual environment.

Assumptions and Considerations

- The USDA Foundation Foods files and the NHANES 2017 to 2018 files are available and are suitable for personal and educational use.
- The NHANES survey weights and strata are not applied, so the reported figures are sample level estimates rather than nationally representative values.
- Diet and diagnosis values in the survey are reported by participants, which introduces the normal effects of self reporting.
- The application is for personal and educational use only and does not provide medical diagnosis or treatment advice.

Risks and Dependencies

- **Data quality:** several survey fields contain missing values. Glucose in particular is measured only on the fasting subsample, so it is missing for about 69 percent of respondents. The pipeline handles this explicitly rather than hiding it.

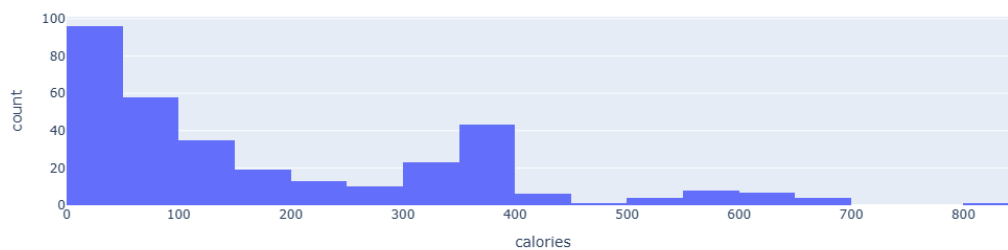
- **Class imbalance:** only about 10 percent of labeled respondents report diabetes. This shapes how the Phase 2 model will be evaluated, with attention to precision, recall, and the area under the ROC curve rather than raw accuracy.
- **Library dependencies:** the solution relies on third party Python libraries that must be installed from the requirements file.
- **Scope risk:** Phase 1 is descriptive by design, and predictive work is deferred to Phase 2.

Appendix A. Data and Analysis Results

The figures in this appendix are the actual charts produced by the Phase 1 analysis notebook. They are grouped into the nutrition analysis of the USDA food catalog and the diabetes insights drawn from the NHANES survey. The diabetes results are associations observed in the sample and should not be read as proof of cause.

Nutrition analysis

1. Calories Distribution



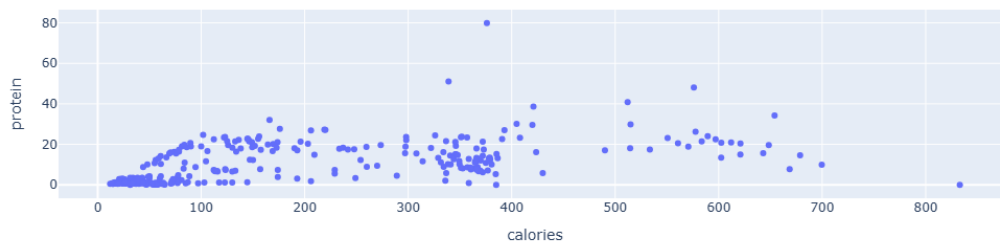
Calorie distribution across the USDA food catalog.

2. Mean Macronutrient Distribution



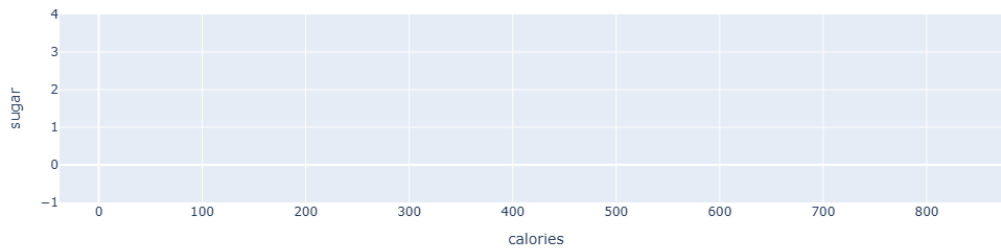
Mean macronutrient distribution.

3. Protein vs Calories



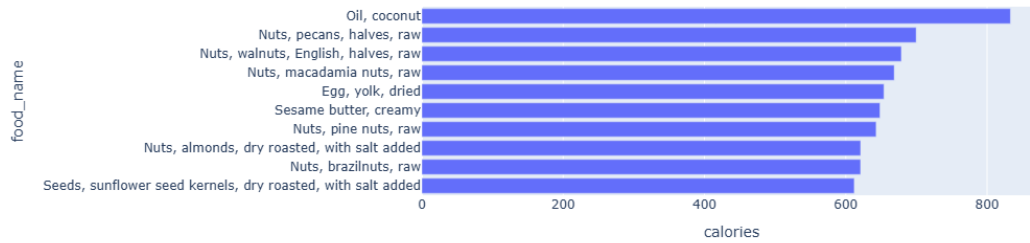
Protein content versus calories.

4. Sugar vs Calories



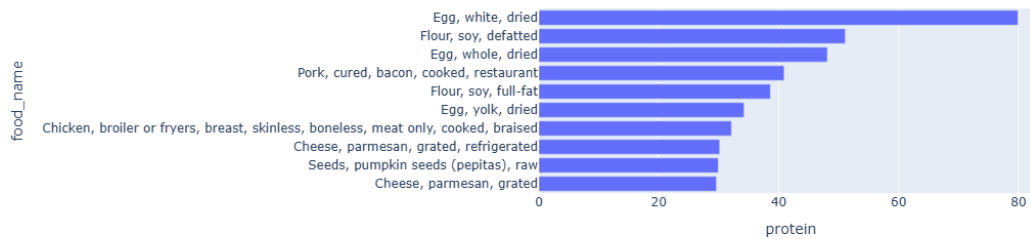
Sugar content versus calories.

10. Top 10 High-Calorie Foods



Top ten high calorie foods.

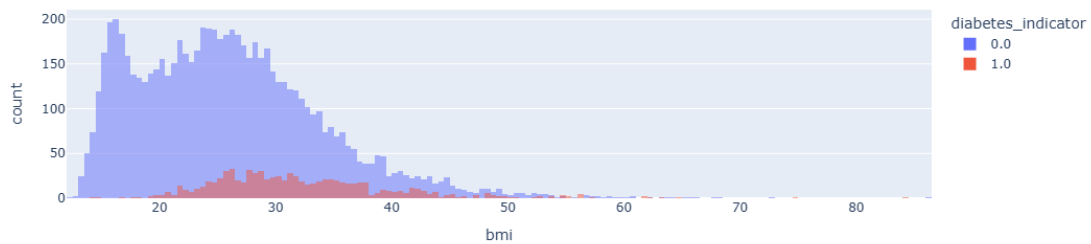
11. Top 10 High-Protein Foods



Top ten high protein foods.

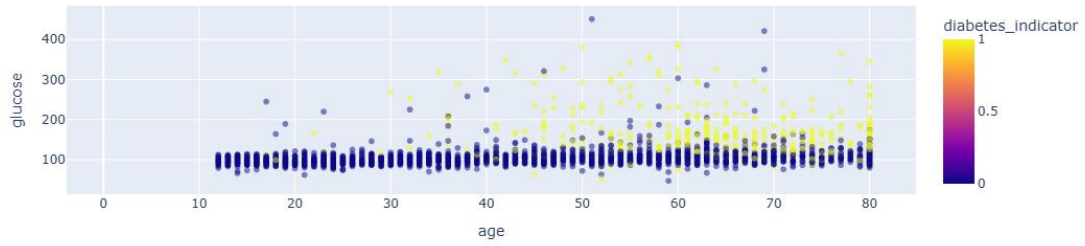
Diabetes insights

5. BMI Distribution



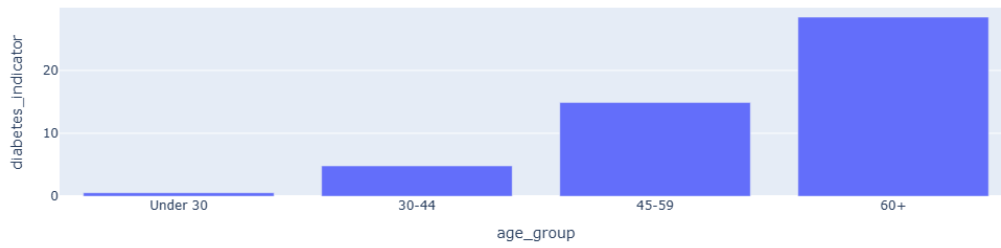
BMI distribution by diabetes status.

6. Age vs Glucose



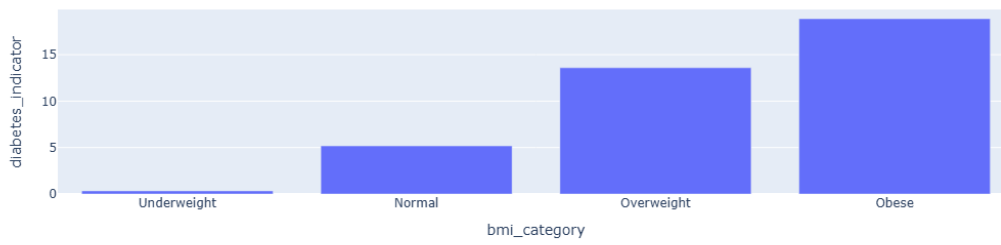
Age versus glucose by diabetes status.

7. Diabetes by Age Group (%)



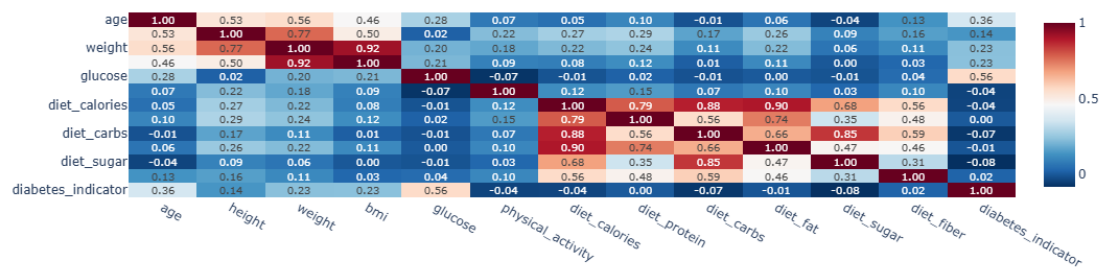
Diabetes prevalence by age group.

8. Diabetes by BMI Category (%)



Diabetes prevalence by BMI category.

9. Correlation Heatmap



Correlation of health and diet features.